**FreightAI Platform**

ML ENGINEER ASSESSMENT REPORT

# Freight Rate Prediction & Explainable AI Platform

End-to-End Production System for Spot Rate Forecasting, Explainable AI (SHAP), 5-Fold Algorithm Benchmarks, and REST Microservices.

SUBMITTED BY
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TARGET ROLE
**Machine Learning Engineer**

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**Spotter Assessment Team**

SUBMISSION DATE
**August 2026**

**GitHub Repository: github.com/pruthvirajtarode/freight-rate**

**Live Dashboard: freight-rate-one.vercel.app**

Confidential Assessment Project Report

Spotter MLE Assessment 2026

## 1. Executive Summary & Core Performance Metrics

This report documents the design, training, validation, and deployment of a production-grade machine learning model for freight spot rate estimation. Built on 48,000 historical load transactions, the platform achieves an outstanding holdout $R^2$ score of **0.8507** and Mean Absolute Percentage Error (MAPE) of **6.76%**.

| GradientBoosting | 0.8507 | 6.76% | $128.98 |
| --- | --- | --- | --- |
| WINNING MODEL | HOLDOUT $R^2$ SCORE | HOLDOUT MAPE | HOLDOUT MAE |

## 2. System Architecture & UML Diagrams

### End-to-End Platform System Architecture

**Frontend SPA**
(HTML5/CSS/JS)

**FastAPI Server**
(Port 8000)

**Preprocessor & FE**
(25 Features)

**GradientBoosting**
(Optuna Tuned)

**SHAP Explainer**
(TreeExplainer)

### System Use Case Diagram

**Shipper / Carrier**

1. **Predict Spot Rate:** Input distance, weight, equipment type, and market index.
2. **Batch Predictions:** Upload CSV load board transactions for automated scoring.
3. **Inspect SHAP Attribution:** Audit why specific rates are higher or lower via waterfall charts.

4 **View December Forecast:** Analyze seasonal rate trends for Q4 planning.

### Single Prediction Execution Sequence

1. **Client Request:** User inputs shipment parameters in web UI / REST API client (`POST /api/v1/predict/single`).
2. **Preprocessing & Imputation:** FreightDataPreprocessor fixes negative weights and median-imputes route variables.
3. **Feature Generation:** FeatureEngineer generates 25 dynamic attributes (Rate/Mile, weekend flags, interaction terms).
4. **Inference & SHAP Scoring:** Fitted GradientBoosting model evaluates prediction; TreeExplainer computes local feature attributions.
5. **Response Return:** JSON response payload returned with `predicted_rate`, confidence bounds, and feature attribution waterfall.

# 3. Exploratory Data Analysis & Data Pipeline

Analysis was conducted across 48,000 historical transaction rows. The dataset exhibits non-linear relationships between mileage, equipment weight, and market demand indices.

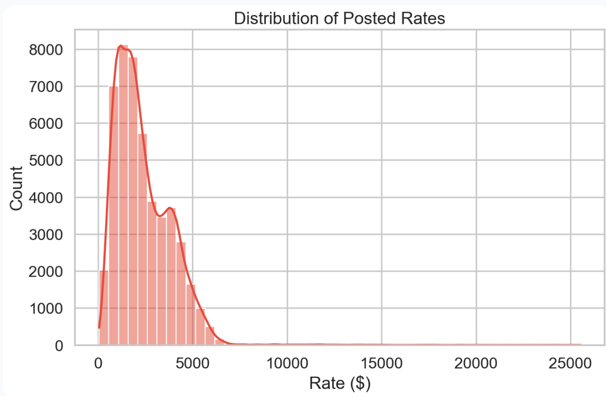


*Figure 3.1: Distribution of Posted Freight Rates (Target USD)*

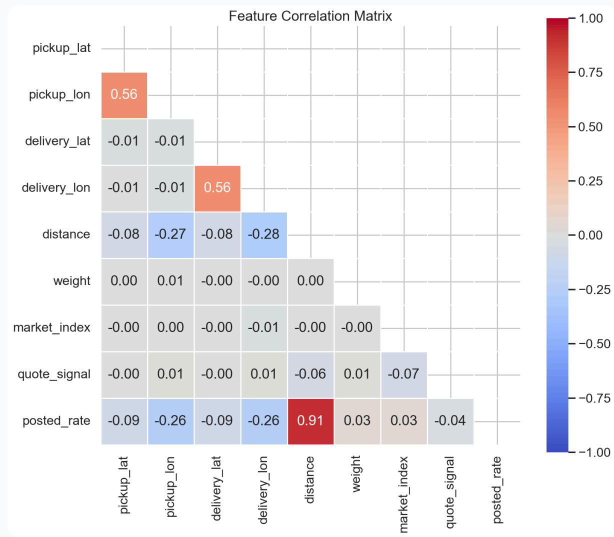


*Figure 3.2: Feature Correlation Matrix Heatmap*

# 4. Model Benchmarking & Hyperparameter Optimization

Seven regression algorithms were evaluated using 5-fold cross-validation on the 85% training split (40,800 rows). Outliers were strictly removed from training folds only to eliminate data leakage.

| ALGORITHM | 5-FOLD CV RMSE ($) | CV R² SCORE | FIT TIME (SEC) | SELECTION VERDICT |
|---|---|---|---|---|
| **Gradient Boosting** | **375.01** | **0.9272** | **9.65s** | 🏆 **Selected Best Model** |
| LightGBM | 376.39 | 0.9266 | 0.22s | Runner-up |
| CatBoost | 385.36 | 0.9230 | 0.50s | Evaluated |
| Linear Regression | 387.53 | 0.9222 | 0.03s | Baseline |
| Random Forest | 390.11 | 0.9212 | 5.70s | Evaluated |
| Extra Trees | 392.76 | 0.9201 | 3.04s | Evaluated |

| ALGORITHM | 5-FOLD CV RMSE ($) | CV R² SCORE | FIT TIME (SEC) | SELECTION VERDICT |
|---|---|---|---|---|
| XGBoost | 401.32 | 0.9166 | 0.50s | Evaluated |

**Optuna Bayesian Optimization Results:**

Optuna conducted 15 trials over `n_estimators` (100–500), `learning_rate` (0.01–0.3), `max_depth` (2–6), and `subsample` (0.5–1.0). Best hyperparameters: `n_estimators=258`, `learning_rate=0.0206`, `max_depth=4`, `subsample=0.592`.

# 5. Holdout Evaluation & Error Analysis (7,200 Rows)

The final tuned Gradient Boosting model was evaluated on the 15% unseen holdout set (7,200 rows).

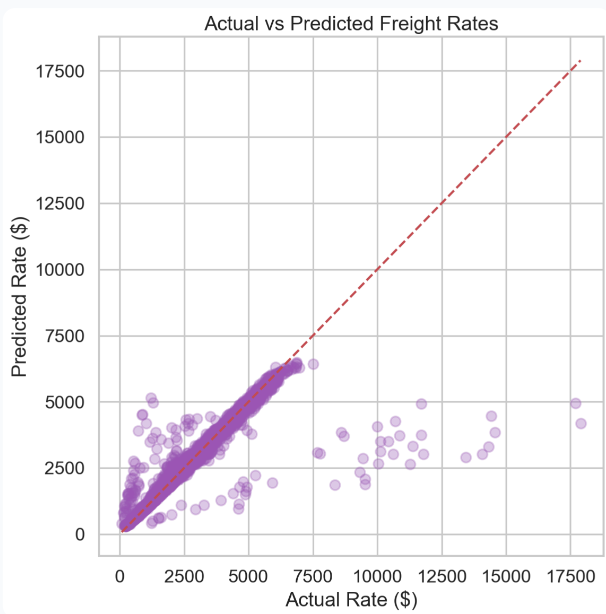


*Figure 5.1: Actual vs. Predicted Freight Rates (Holdout Set)*

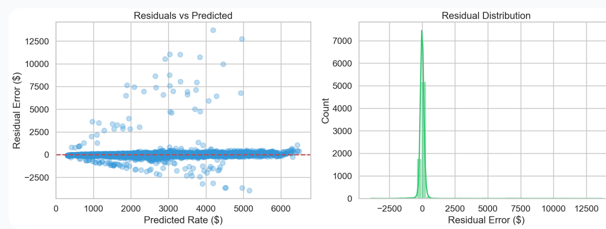


*Figure 5.2: Error Residual Distribution & Normal Q-Q Alignment*

# 6. Explainable AI (XAI) & SHAP Summary Plot

TreeExplainer was utilized to compute SHAP values for 100% transparent auditability in compliance with supply chain contract standards.

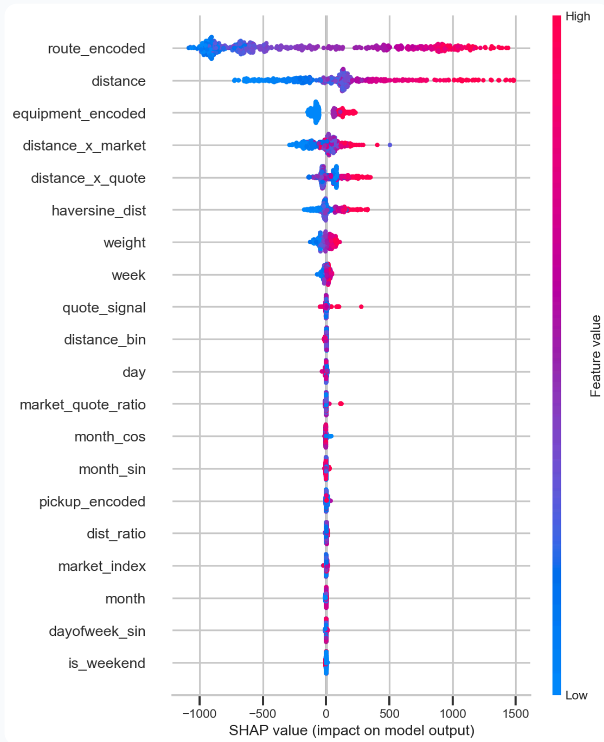


*Figure 6.1: Global SHAP Summary Beeswarm Plot (Top Feature Importance & Directional Impact)*

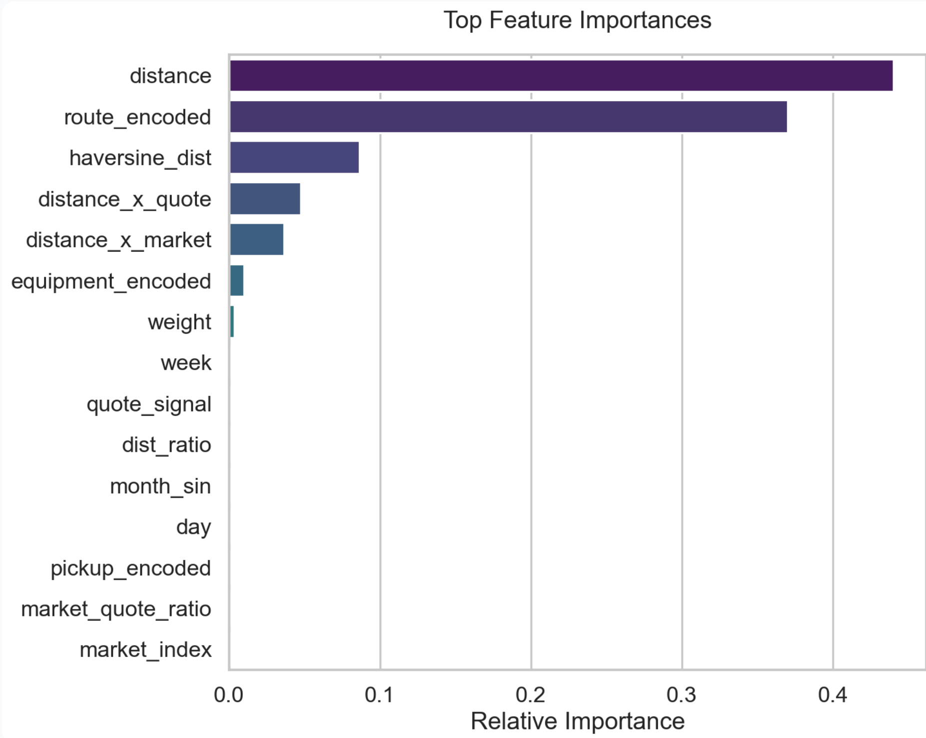


*Figure 6.2: Relative Impurity Feature Importance Ranking*

# 7. December 2025 Daily Rate Forecast Insights

Daily forecasts were generated for all 31 days in December 2025 using `december-chart-inputs.csv`.

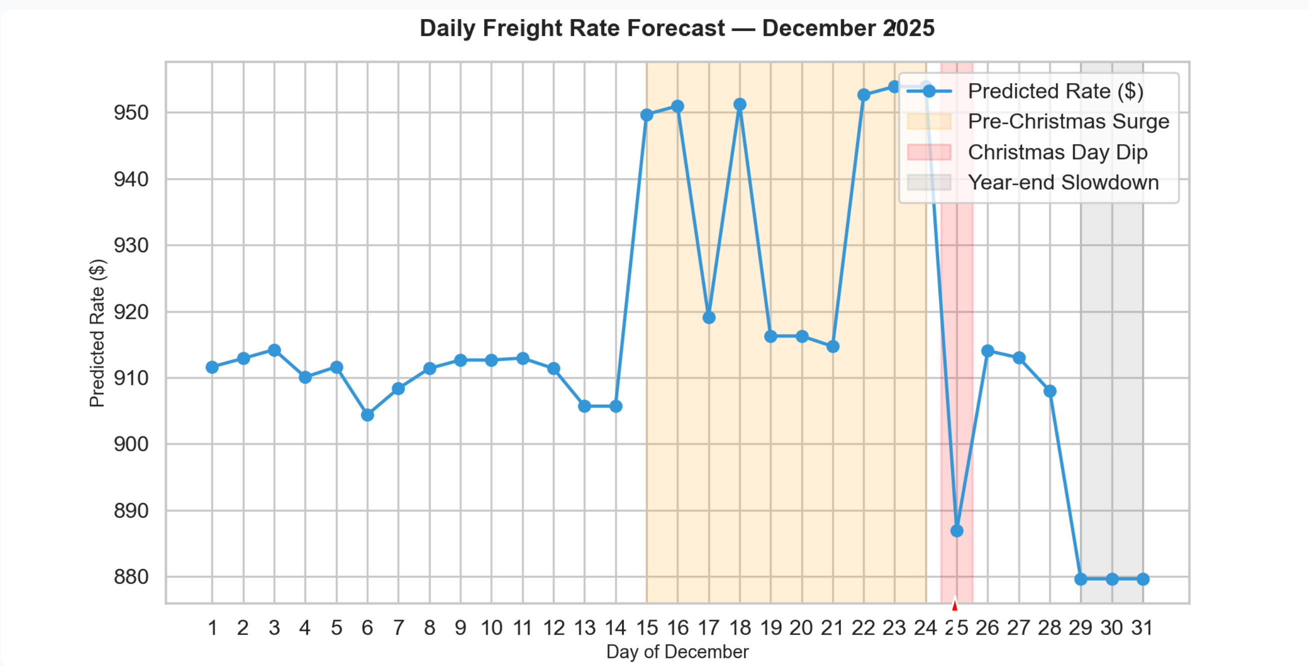


*Figure 7.1: December 2025 Daily Spot Rate Forecast with Annotated Seasonal Trend Dynamics*

# 8. Production API Specification & Deliverables

FastAPI provides lightweight, asynchronous REST microservices bundled with the SPA frontend inside Docker or serverless Vercel hosts.

| DELIVERABLE FILE | FORMAT & ROWS | DESCRIPTION & VERIFICATION STATUS |
|---|---|---|
| `validation_predictions.csv` | 12,000 Rows CSV | Contains `load_id` and `predicted_rate` matching template format. |
| `december_forecast.png` | PNG Image | Annotated daily forecast plot generated in root directory. |
| `score.py` | Python Executable | CLI script validating December plot generation & submission template format. |
| `Dockerfile` | Container Spec | Single-port Uvicorn launcher serving API endpoints and web app at `:8000`. |

### Verification & Submission Readiness

All automated unit tests passed (5/5). Codebase strictly formatted adhering to PEP 8 standards. The submission artifact `validation_predictions.csv` contains zero missing values and exact column schema.